

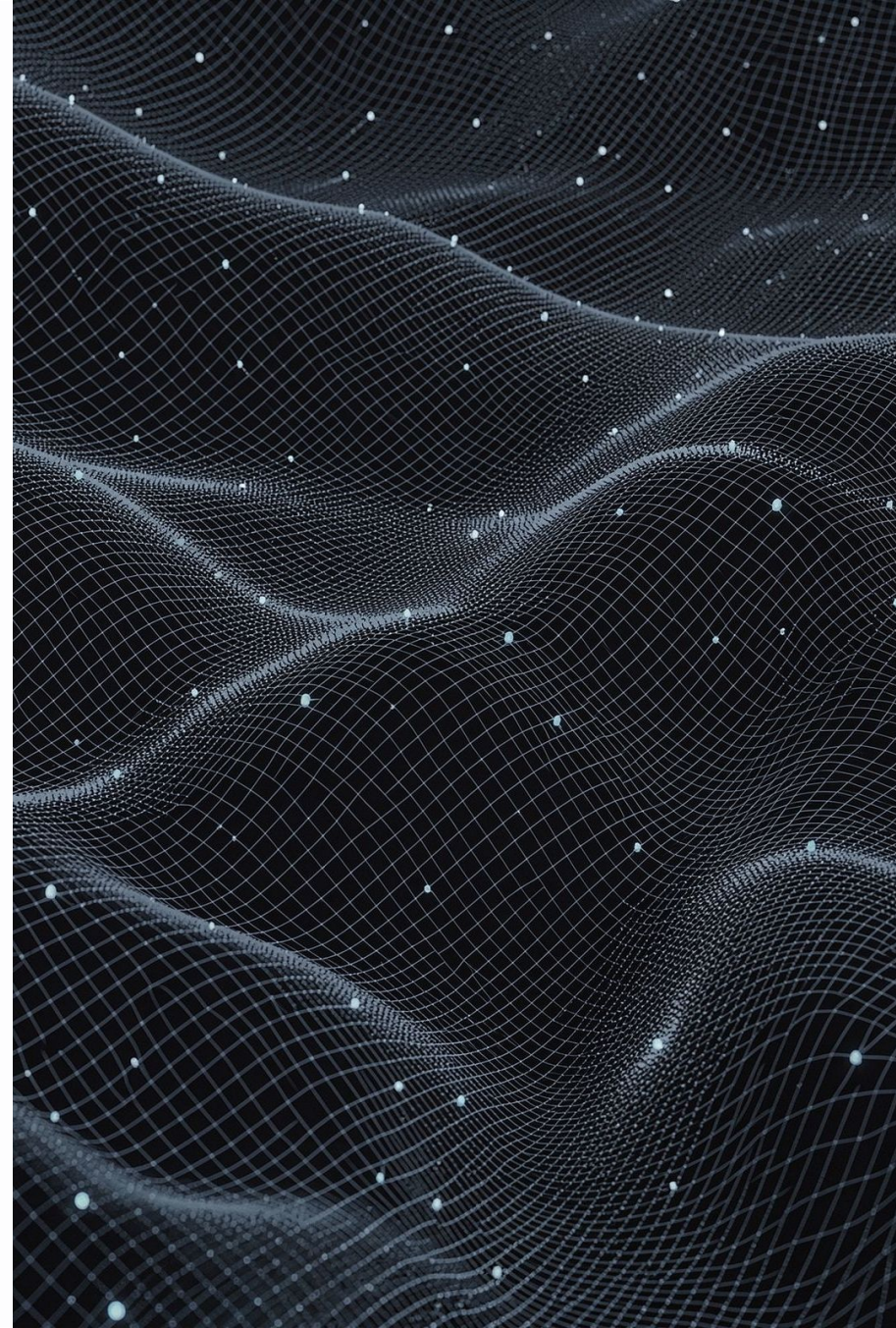
Machine Learning Internship at Pratinik InfoTech

June – August 2026

SHUBHAM KUMAR

CS-2341589

2023-27



About Pratinik InfoTech

AI & ML Solutions

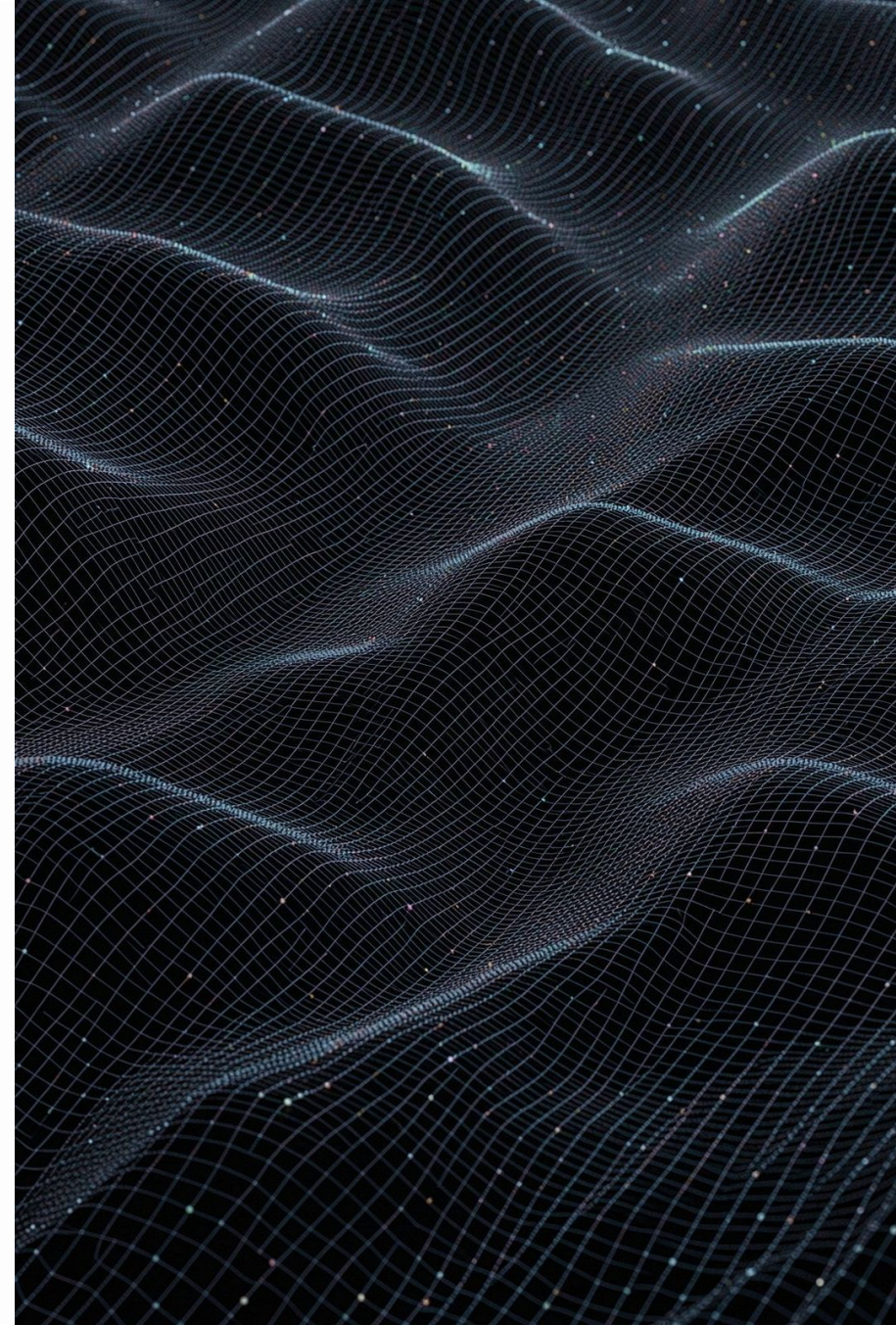
Leading provider of enterprise-grade artificial intelligence and machine learning implementations

Enterprise Focus

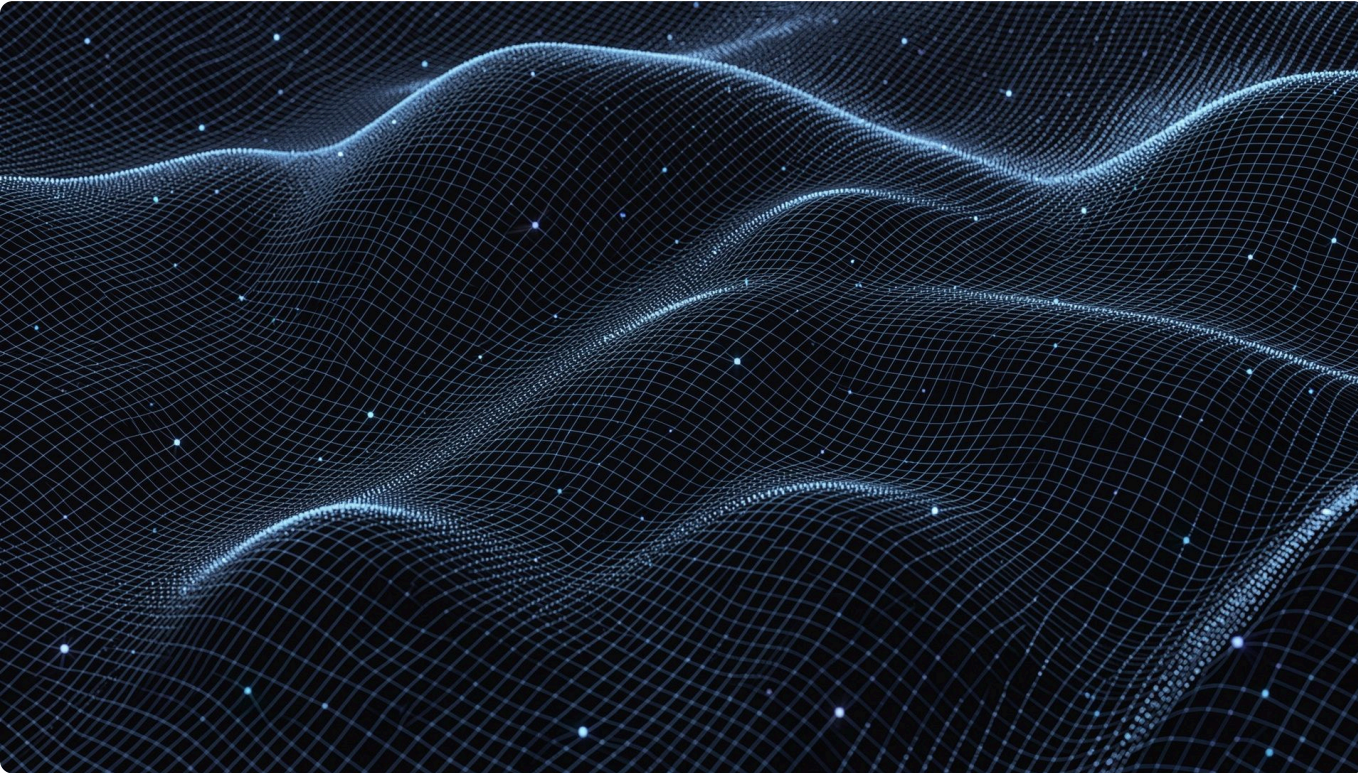
Specializes in building robust, scalable ML systems for real-world industry challenges

Mentorship-Driven

Internship program structured around guided learning, peer collaboration, and professional growth



Internship Overview



Duration

10 June – 9 August 2026 | 2 months
of intensive ML development

Focus Area

Machine learning development and
deployment – end-to-end model
lifecycle

Mentor

Muskaan Sharma, HR Executive,
Pratinik InfoTech

Core ML Competencies Developed



Supervised & Unsupervised Learning

Applied classification, regression, and clustering algorithms to structured datasets, gaining fluency in both paradigms



Data Preprocessing & Feature Engineering

Cleaned, transformed, and enriched raw data to prepare high-quality feature sets for model training



Model Evaluation & Optimization

Used cross-validation, confusion matrices, and grid search to measure and improve model accuracy and generalization

Technical Skills Gained

Hands-on proficiency with the core tools and frameworks that power modern machine learning engineering.



1 Python for Machine Learning

scikit-learn for classical ML, TensorFlow for deep learning architectures

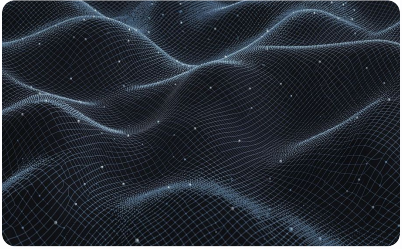
2 Data Analysis & Visualization

Pandas, NumPy, Matplotlib, and Seaborn for exploratory data analysis and insight communication

3 Model Deployment & Pipelines

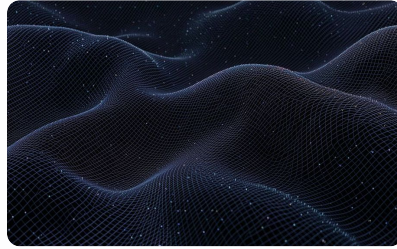
Production-ready pipelines using Flask APIs and containerized deployment workflows

Key Projects & Applications



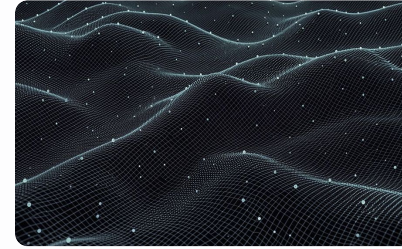
Classification Models

Built and validated classification models on real-world datasets, including binary and multi-class scenarios using decision trees, SVM, and ensemble methods



Predictive Analytics

Implemented regression and time-series forecasting pipelines to extract actionable insights from historical enterprise data



End-to-End ML Workflow

Executed complete ML workflows — from raw data ingestion and feature selection through training, evaluation, and production deployment

Hands-On Experience

Industry-Standard Datasets

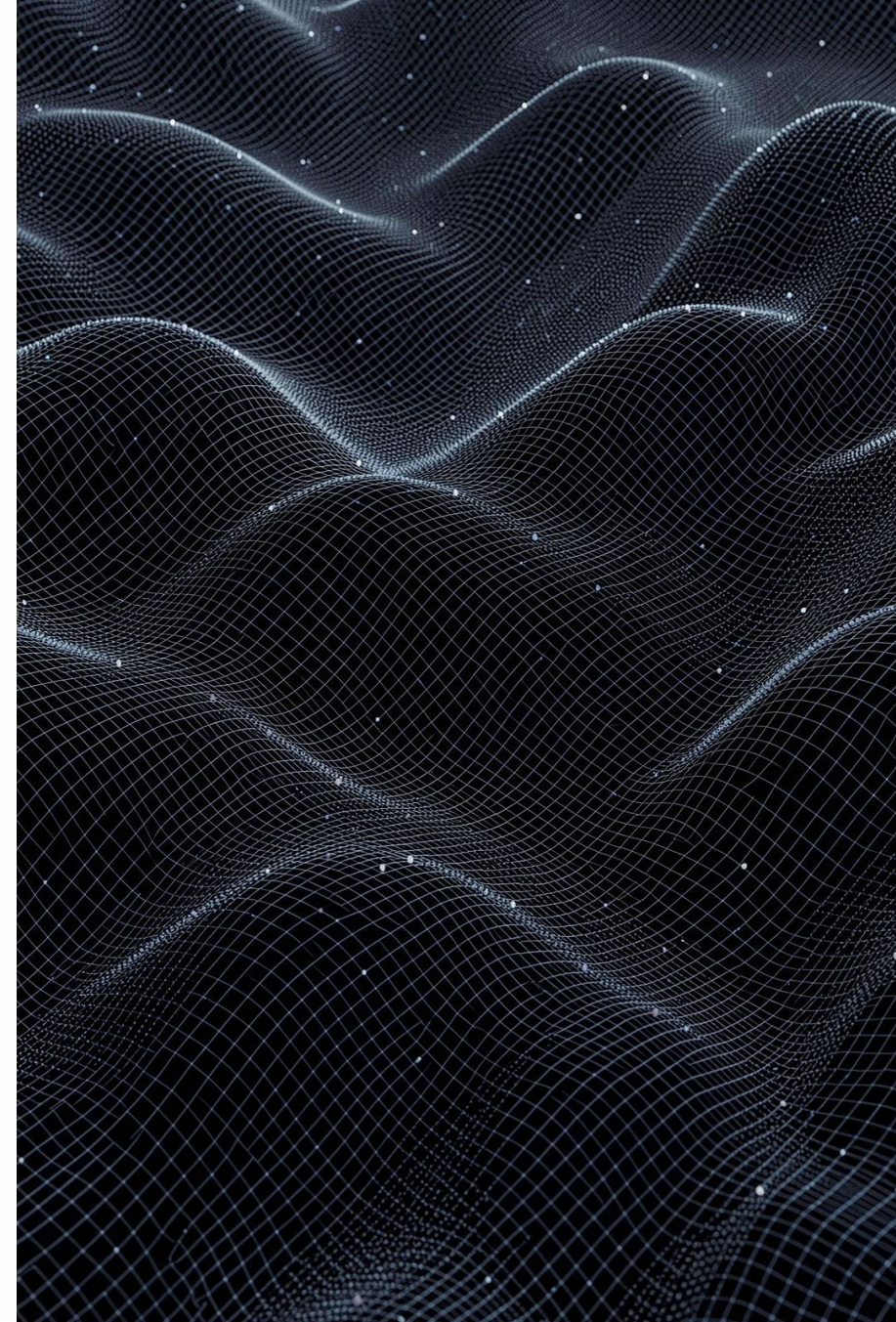
Worked with real enterprise datasets, navigating issues of scale, noise, and missing values in professional settings

Senior Engineer Collaboration

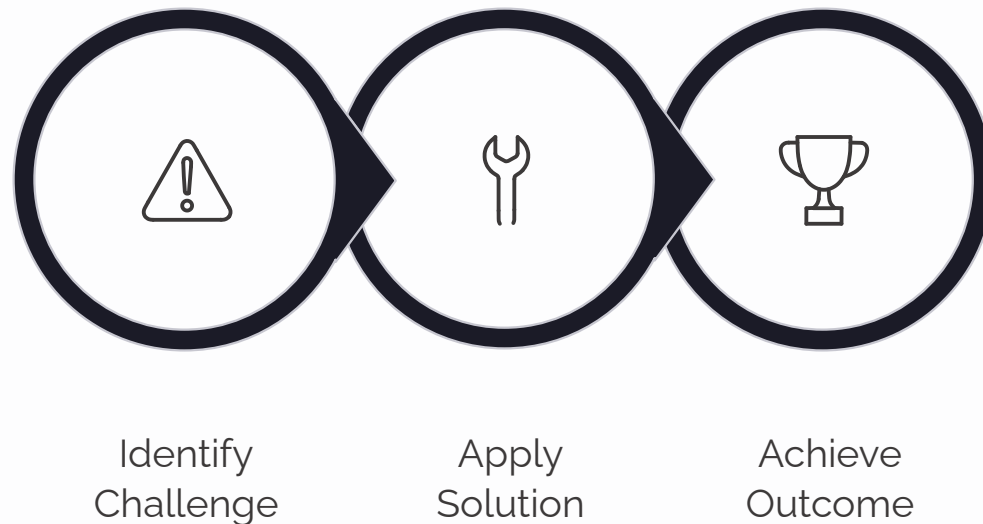
Paired with senior ML engineers for code reviews, architecture discussions, and iterative model refinement

Best Practices in ML Development

Adopted industry conventions including version control, reproducible experiments, and modular code structure



Challenges & Solutions



→ Imbalanced Datasets

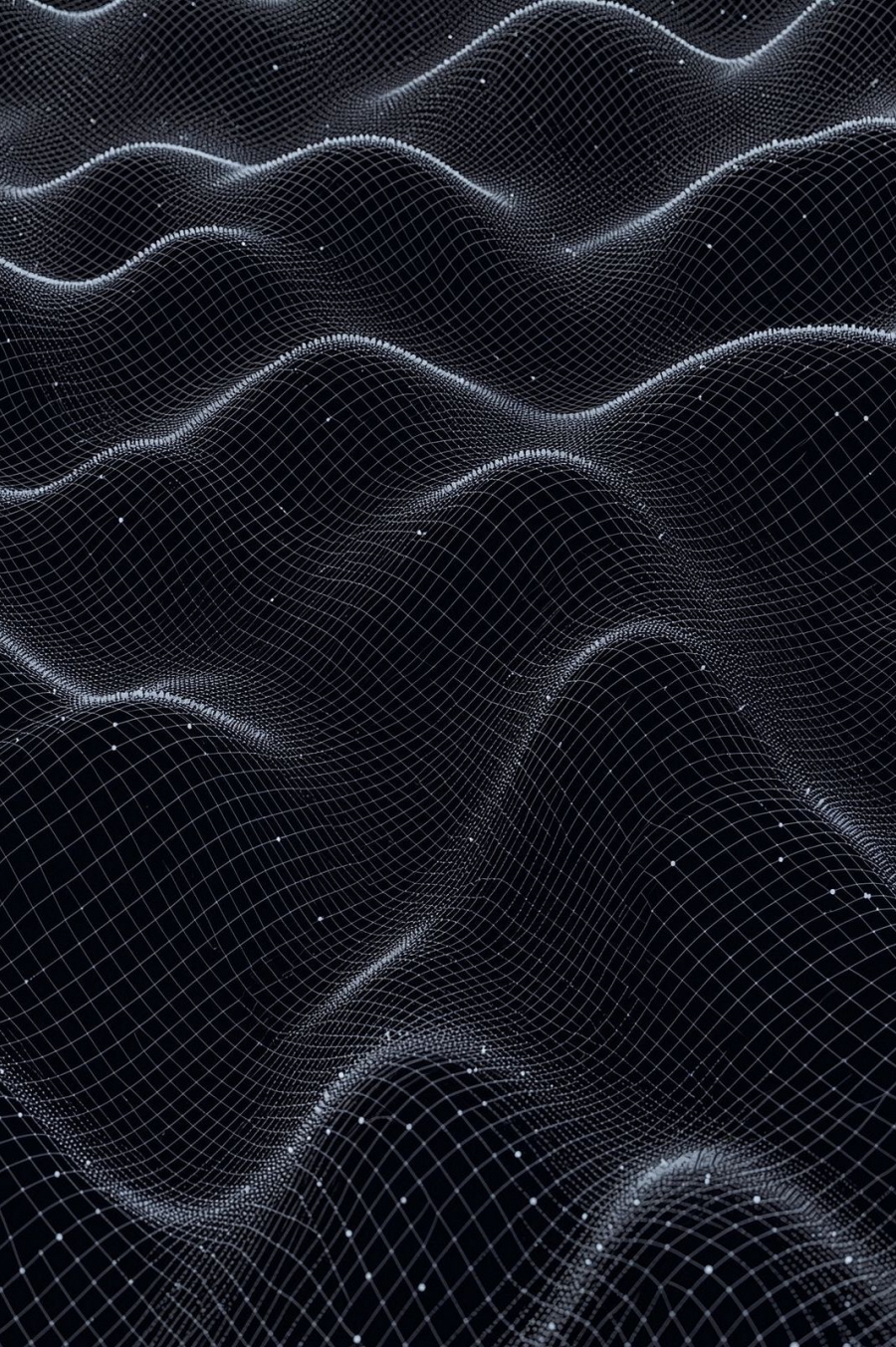
Applied SMOTE and undersampling resampling techniques to restore class balance and improve minority-class recall

→ Model Performance Gaps

Conducted systematic hyperparameter tuning using GridSearchCV and RandomizedSearchCV to maximize evaluation metrics

→ Production Deployment

Overcame integration challenges to successfully deploy trained models via REST APIs in live production environments



Key Takeaways

ML Lifecycle Mastery

Developed a deep, end-to-end understanding of the machine learning lifecycle – from raw data collection and preprocessing through model training, evaluation, and production deployment

Enterprise Tools & Frameworks

Gained practical, hands-on experience with industry-standard ML tools, frameworks, and workflows used in real enterprise environments

Professional Development

Strengthened collaborative and professional skills by working alongside experienced engineers in a mentorship-rich, team-oriented tech environment